

Assignment 14

Problem Statement:-Create an Elastic IP for an instance.

Step1:First we will create an instance. Select ubuntu, key pair, select all the check box in security.

It seems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices.

Do not show me this message again Take a walkthrough

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name: rs Add additional tags

Application and OS Images (Amazon Machine Image)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below.

Search your full catalog including 1000s of application and OS images

Quick Start

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux Debian

Browse more AMIs

Summary

Number of instances: 1

Software Image (AMI): Canonical, Ubuntu, 24.04, amd64...read more

Virtual server type (instance type): t2.micro

Firewall (security group): New security group

Storage (volumes): 1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

Subnet

No preference (Default subnet in any availability zone)

Auto-assign public IP

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups)

Create security group Select existing security group

We'll create a new security group called 'launch-wizard-1' with the following rules:

- Allow SSH traffic from: Anywhere (0.0.0.0/0)
- Allow HTTPS traffic from the internet: To set up an endpoint, for example when creating a web server
- Allow HTTP traffic from the internet: To set up an endpoint, for example when creating a web server

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Configure storage

1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

Cancel Launch instance Preview code

Step2:Now select instance and stop instance.After some time start instance. Note both pervious and now public IPv4 address.

AWS Management Console - EC2 Instances

Instances (1) info

Find Instance by attribute or tag (case-sensitive)

Instance ID = i-0166b97addad2f6f8

Clear filters

All states

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
rs	i-0166b97addad2f6f8	Running	t2.micro	Initializing	View alarms +	us-east-1a	ec2-100-27-29-86.com...

Select an instance

AWS Management Console - EC2 Instances

Successfully initiated stopping of i-0166b97addad2f6f8

Instances (1/1) info

Find Instance by attribute or tag (case-sensitive)

Instance ID = i-0166b97addad2f6f8

Clear filters

All states

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
rs	i-0166b97addad2f6f8	Stopping	t2.micro	-	View alarms +	us-east-1a	ec2-100-27-29-86.com...

i-0166b97addad2f6f8 (rs)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

Instance summary info

Instance ID: i-0166b97addad2f6f8

IPV6 address: -

Hostname type: -

Public IPv4 address: 100.27.29.86 | open address

Instance state: Stopping

Private IPv4 addresses: 172.31.90.141

Public IPv4 DNS: ec2-100-27-29-86.compute-1.amazonaws.com | open address

Private IP DNS name (IPv4 only): -

Step3:Go to Elastic IPs and allocate it.

AWS Management Console - Elastic IP addresses

Elastic IP address allocated successfully. Elastic IP address 34.233.96.141

Associate this Elastic IP address

Elastic IP addresses (1)

Find elastic IP addresses by attribute or tag

Public IPv4 address : 34.233.96.141

Actions | Allocate Elastic IP address

Name	Allocated IPv4 address	Type	Allocation ID	Reverse DNS record	Associated instance
-	34.233.96.141	Public IP	eipalloc-09d8f26c1c103f141	-	-

View IP address usage and recommendations to release unused IPs with Public IP insights.

Step4:Now associate Elastic IP address, select resource type instances.

Associate Elastic IP address
Choose the instance or network interface to associate to this Elastic IP address (34.233.96.141)

Elastic IP address: 34.233.96.141

Resource type
Choose the type of resource with which to associate the Elastic IP address.

☒ Instance
☐ Network interface

Warning: If you associate an Elastic IP address with an instance that already has an Elastic IP address associated, the previously associated Elastic IP address will be disassociated, but the address will still be allocated to your account. [Learn more](#)

If no private IP address is specified, the Elastic IP address will be associated with the primary private IP address.

Instance
i-0166b97addad2f6f8

Private IP address
The private IP address with which to associate the Elastic IP address.
172.31.90.141

Reassociation
Specify whether the Elastic IP address can be reassigned with a different resource if it already associated with a resource.
☒ Allow this Elastic IP address to be reassigned

[Cancel](#) [Associate](#)

Step5:Again now stop and start the instance, no IPv4 address will be changed. (before stopping- 3.216.220.105, after stopping- 3.216.220.105)

Instances (1/1)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
rs	i-0166b97addad2f6f8	Running	t2.micro	2/2 checks passed	View alarms	us-east-1a	ec2-34-233-96-141.co...

i-0166b97addad2f6f8 (rs)

Instance summary

Instance ID
i-0166b97addad2f6f8

IPv6 address
-

Hostname type
IP name: ip-172-31-90-141.ec2.internal

Answer private resource DNS name
IPv4 (A)

Auto-assigned IP address
-

IAM Role
-

IMDSv2
Required

Operator
-

Public IPv4 address
34.233.96.141 | [open address](#)

Instance state
Running

Private IP DNS name (IPv4 only)
ip-172-31-90-141.ec2.internal

Instance type
t2.micro

VPC ID
vpc-0975da0bd01b6150d

Subnet ID
subnet-0032e9b662f71344b

Instance ARN
arn:aws:ec2:us-east-1:954976318775:instance/i-0166b97addad2f6f8

Private IPv4 addresses
172.31.90.141

Public IPv4 DNS
ec2-34-233-96-141.compute-1.amazonaws.com | [open address](#)

Elastic IP addresses
34.233.96.141 [Public IP]

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | [Learn more](#)

Auto Scaling Group name
-

Managed
false